# An MPI Cell-Centered Finite-Volume Solver for the 2-D Compressible Navier–Stokes Equations: NACA0012 and Cylinder Benchmark Cases

CFD Solver Agentic Benchmark Submission

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**Abstract**

This report documents `cfd_solver`, an original C++17/MPI cell-centered finite-volume solver for the two-dimensional compressible Navier–Stokes equations of a calorically perfect gas. The solver combines a second-order weighted least-squares reconstruction with a Barth–Jespersen limiter and positivity fallback, a Roe approximate Riemann solver with a Harten–Yee entropy fix, viscous stress and Fourier heat-flux terms for laminar flow, an implicit LU-SGS pseudo-time march with local CFL control, and a true BDF2 physical-time outer loop with inner iterations for transient cases. The mesh is partitioned with METIS and all halo data are exchanged with neighbor-scoped MPI point-to-point communication; no full mesh or full conservative state is replicated during solver iterations.

All eight required cases were completed successfully. The six NACA0012 cases (inviscid and laminar $Re = 5000$ at $M_\infty = 0.15$, 0.8, and 2.0) converged to a target reduction or a documented stable-force plateau, the circular-cylinder $Re = 20$ case converged to a steady laminar wake with $C_D = 2.07$, and the $Re = 200$ cylinder case reached a statistically periodic vortex street over the full $t_f = 300$ horizon with a shedding Strouhal number $St \approx 0.055$.

## 1 Introduction

The benchmark objective is to build a complete unstructured finite-volume CFD solver that runs the supplied NACA0012 and cylinder cases to converged or statistically settled results, writes machine-readable residual and force histories, produces flow-field files and visualizations, and documents the work in an academic-style report. The eight required cases are

- NACA0012 at zero angle of attack, inviscid, at $M_\infty = 0.15$, 0.8, and 2.0;
- NACA0012 at zero angle of attack, laminar $Re = 5000$, at the same three Mach numbers;
- circular cylinder, laminar $Re = 20$, steady wake;
- circular cylinder, laminar $Re = 200$, unsteady vortex shedding with $\Delta t = 0.01$ to $t_f = 300$.

The report is organized as follows. Sections 2–4 give the governing equations, meshes, and spatial discretization; Section 5 describes boundary conditions; Sections 6–7 describe the implicit/transient time integration and the MPI strategy; Section 8 describes outputs, reproducibility, and sanity checks; Sections 9–10 present the results and MPI rank-count validation; Sections 11–13 cover visualization conventions and honest limitations. All figures are stored in `report/figures/` and mapped to their source files in `report/figure_manifest.csv`.

## 2 Governing equations and nondimensionalization

The conservative form of the 2-D compressible Navier–Stokes equations is

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}^i}{\partial x} + \frac{\partial \mathbf{G}^i}{\partial y} = \frac{\partial \mathbf{F}^v}{\partial x} + \frac{\partial \mathbf{G}^v}{\partial y}, \tag{1}$$

with the conservative state $\mathbf{U} = [\rho, \rho u, \rho v, \rho E]^T$, inviscid fluxes

$$\mathbf{F}^i = [\rho u, \rho u^2 + p, \rho uv, u(\rho E + p)]^T, \qquad \mathbf{G}^i = [\rho v, \rho uv, \rho v^2 + p, v(\rho E + p)]^T, \tag{2}$$

and viscous fluxes built from the Newtonian stress tensor

$$\tau_{xx} = 2\mu u_x - \frac{2}{3}\mu(u_x + v_y), \quad \tau_{yy} = 2\mu v_y - \frac{2}{3}\mu(u_x + v_y), \quad \tau_{xy} = \mu(u_y + v_x), \tag{3}$$

and Fourier heat flux $\mathbf{q} = -k\nabla T$, $k = \mu c_p/Pr$.

The gas is calorically perfect: $p = (\gamma - 1)\rho e$, $E = e + \frac{1}{2}(u^2 + v^2)$, and $a = \sqrt{\gamma p/\rho}$, with $\gamma = 1.4$, $R = 1$, and $Pr = 0.72$ in all cases. All quantities are nondimensional. The case files specify $\rho_\infty = 1$, $|\mathbf{V}_\infty| = 1$, a reference length $L = 1$ (airfoil chord or cylinder diameter), and the freestream pressure $p_\infty = 1/(\gamma M_\infty^2)$, which corresponds to the supplied values (e.g. 31.746 at $M_\infty = 0.15$). The dynamic pressure is $q_\infty = \frac{1}{2}\rho_\infty V_\infty^2 = 0.5$, and force and moment coefficients are

$$C_D = \frac{F_D}{q_\infty L}, \qquad C_L = \frac{F_L}{q_\infty L}, \qquad C_m = \frac{M}{q_\infty L^2}, \tag{4}$$

with the moment evaluated about $(0.25, 0)$ for the airfoil and $(0, 0)$ for the cylinder. For laminar cases the constant viscosity is set from the case Reynolds number,

$$\mu = \frac{\rho_\infty V_\infty L}{Re}, \tag{5}$$

which matches $Re = 20$, $Re = 200$, and $Re = 5000$ exactly.

## 3 Meshes, boundary tags, and case inputs

Both meshes are imported from CGNS. `mesh.cpp` reads unstructured zones, cell connectivity, face geometry, and boundary families, and builds cell volumes, face areas, outward face normals, centroids, and wall edge lengths. Multi-zone files with 1-to-1 abutting interfaces are matched by edge coordinates and merged into a single cell/face graph before partitioning. Boundary families are mapped from the case-file boundary conditions: `bc-2/FAR` become farfield and `bc-4/WALL` become either a slip wall (inviscid cases) or a no-slip adiabatic wall (laminar cases). Table 1 summarizes the two meshes.

| Mesh | Cells | Faces | Wall rows | Wall BC | Farfield BC |
|---|---|---|---|---|---|
| `NACA0012_H2.cgns` | 20,816 | 36,498 | 404 | slip or no-slip | farfield |
| `CylinderB1.cgns` | 10,185 | 20,180 | 100 | no-slip adiabatic | farfield |

Table 1: Mesh summary: cell/face counts are the global values used in every run; wall rows are the boundary rows written to `surface.csv`.

Table 2 lists the supplied production run controls. The solver parses every case entirely from its JSON file and uses these values without modification; no source edit is needed between cases.

| Case | Type | $M_\infty$ | $Re$ | $N_{\max}$ | $\mathrm{CFL}_0 \to \mathrm{CFL}_{\max}$ ($N_{\mathrm{ramp}}$) | Inner $n_{\min}/n_{\max}$ (target) | Outer target |
|---|---|---|---|---|---|---|---|
| NACA M0.15 inviscid | steady | 0.15 | — | 20,000 | $1.0 \to 100$ (2000) | 3/50 ($10^{-2}$) | 4.0 |
| NACA M0.8 inviscid | steady | 0.80 | — | 30,000 | $1.0 \to 100$ (3000) | 3/50 ($10^{-2}$) | 4.0 |
| NACA M2.0 inviscid | steady | 2.00 | — | 40,000 | $0.2 \to 50$ (5000) | 3/80 ($10^{-2}$) | 3.0 |
| NACA M0.15 laminar | steady | 0.15 | 5000 | 40,000 | $1.0 \to 100$ (5000) | 3/80 ($10^{-2}$) | 4.0 |
| NACA M0.8 laminar | steady | 0.80 | 5000 | 40,000 | $0.5 \to 80$ (5000) | 3/80 ($10^{-2}$) | 4.0 |
| NACA M2.0 laminar | steady | 2.00 | 5000 | 50,000 | $0.2 \to 50$ (7000) | 3/100 ($10^{-2}$) | 3.0 |
| Cylinder Re20 | steady | 0.10 | 20 | 30,000 | $1.0 \to 100$ (3000) | 3/80 ($10^{-2}$) | 5.0 |
| Cylinder Re200 | transient | 0.10 | 200 | 30,000 ($\Delta t$=0.01, $t_f$=300) | $1.0 \to 1.0$ (0) | 5/1000 ($10^{-3}$) | — |

Table 2: Production run controls taken from the supplied case JSON files without modification. The outer target is the requested residual reduction in orders of magnitude; the Re200 inner target is measured on the total physical-time residual.

# 4 Spatial discretization

The solver is a cell-centered finite-volume method on the unstructured mesh. For every interior face the numerical inviscid flux is evaluated from left/right reconstructed states with the Roe scheme and a quadratic Harten–Yee entropy fix (Rusanov is also implemented and available through an environment switch). Viscous fluxes use face-averaged primitive gradients for velocity and temperature, with the stress tensor and heat flux of Section 2. Boundary faces use the boundary fluxes of Section 5.

## 4.1 Second-order reconstruction

Cell gradients are computed by inverse-distance-weighted least squares over the face-neighbor stencil. The left/right face states are obtained from the piecewise-linear reconstruction

$$\mathbf{W}_f = \mathbf{W}_c + \phi_c \nabla \mathbf{W}_c \cdot (\mathbf{x}_f - \mathbf{x}_c), \tag{6}$$

where $\phi_c \in [0, 1]$ is the Barth–Jespersen limiter value for the cell. Reconstruction is active in all production runs.

## 4.2 Limiter and positivity control

The Barth–Jespersen limiter prevents new local extrema in reconstructed primitive states. As a positivity fallback, if a reconstructed density or pressure is nonphysical at a face, that cell is degraded to a first-order limited state for that step (the face state equals the cell-center state). The number of affected cell-steps is recorded in each `metadata.json` as `positivity_fallback_cell_steps`; the counts are small for the smooth-flow cases (e.g. 26 for NACA M0.15 inviscid, 216 for NACA M0.15 laminar, 0 for the $Re = 20$ cylinder) and are concentrated at shock cells in the $M_\infty = 2.0$ cases (616,858 and 582,139 cell-steps for the inviscid and laminar runs, respectively, over 200–425 pseudo steps on 20,816 cells). No production case is first-order-only; smooth regions remain second order.

## 4.3 Inviscid and viscous fluxes

The inviscid flux is the Roe flux with a quadratic Harten–Yee entropy fix, which is important for the transonic and supersonic cases. The viscous flux uses the Newtonian stress tensor and Fourier heat flux defined in Section 2, with wall-normal gradients projected so that the adiabatic wall condition is imposed exactly. Pressure force is separated from tangential skin-friction force in the force integration: the drag/lift columns in `forces.csv` report $C_{D,p}/C_{D,v}$ and $C_{L,p}/C_{L,v}$ separately, and skin friction uses the tangential wall shear only.

# 5 Boundary conditions

**Farfield.** The farfield boundary is treated with the same approximate Riemann solver used for interior faces, with the exterior state set to the freestream state.

**Inviscid slip wall.** The wall flux contains only the pressure momentum, $F_w = (0, pn_x, pn_y, 0)^T$ with the normal pointing out of the fluid domain; mass and energy fluxes vanish, so the wall is impermeable. Surface rows for slip walls report the boundary values with near-zero normal velocity and nonzero tangential velocity.

**No-slip adiabatic wall.** The wall velocity is zero and the normal component of the temperature gradient is projected to zero, giving zero normal heat flux. The wall momentum flux contains the pressure contribution plus the viscous shear stress evaluated from the wall-cell gradient. Surface rows for no-slip walls report $u = v = M = 0$ (boundary values, not adjacent cell-center values).

# 6 Implicit and transient time integration

## 6.1 Steady pseudo-time march

Steady cases use pseudo-time marching with local CFL-controlled time steps

$$\Delta\tau_i = \mathrm{CFL}\,\frac{V_i}{\sum_f \left[(|u_n| + a)S_f + \frac{\mu}{\rho}\frac{S_f^2}{V_i}\right]}, \tag{7}$$

where the second term is the viscous spectral-radius contribution. The CFL is ramped from the case `cfl_initial` to `cfl_max` over `pseudo_cfl_ramp_steps` (Table 2). The nonlinear system is relaxed with a simplified LU-SGS scheme: for the linear system $(D + L + U)\Delta U = -R$, the diagonal is

$$D_i = \frac{V_i}{\Delta\tau_i} + \frac{1}{2}\sum_f (|u_n| + a)S_f + \alpha V_i, \tag{8}$$

consistent with the $0.5S_f(\Delta F_f - \lambda_f \Delta U_f)$ off-diagonal flux-difference linearization; $\alpha$ is the BDF2 physical-time coefficient (zero for steady runs). Each inner iteration performs forward and backward Gauss–Seidel sweeps, applies the correction with positivity protection, and re-evaluates the nonlinear residual; the inner loop is accepted when the residual ratio falls below `inner_residual_reduction_target` or the iteration cap is reached. Observed min/mean/max inner iterations, target misses, converged-step fraction, and last residual ratio are written to `metadata.json` for every case.

## 6.2 Convergence criterion

Steady cases are marked *converged* when either (i) the residual reduction reaches the case target, or (ii) the forces are stable over the last 100 steps with drift below $T_C = 0.0002 + 0.005|C_D|$ (and $C_L$ drift below $2T_C$) while the residual reduction has reached $\max(1.0, \text{target} - 2.0)$ orders or the absolute residual is below $10^{-5}$. The plateau condition is needed for shock-dominated cases whose relative reduction stalls at 1–2 orders even though the absolute residual and the forces have reached their discrete steady state. The `run_status.json notes` field records which criterion was used.

**Inner-iteration statistics.** The metadata field `inner_target_converged_fraction` counts the fraction of steps in which the inner LU-SGS relaxation reduced its residual ratio below `inner_residual_reducti` In steady runs every pseudo step may legitimately cap the inner-iteration count while the outer residual still decreases, so this fraction can be low or even zero for a fully converged case; the outer criteria above are the authoritative steady-state assessment. For the Re=200 transient case the same statistic is the accepted-step rate and is reported as required (it is 1.0 in the submitted run).

### 6.3 Transient BDF2 loop for cylinder Re 200

Transient cases use BDF2 with a true physical-time outer loop,

$$\frac{V}{\Delta t}\left(\frac{3}{2}U^{n+1} - 2U^{n} + \frac{1}{2}U^{n-1}\right) + R(U^{n+1}) = 0, \tag{9}$$

with backward Euler on the first step. At each physical step the inner relaxation loop solves for $U^{n+1}$ while $U^{n}$ and $U^{n-1}$ remain frozen; the physical-time histories are updated only after the inner solve is accepted. The supplied production settings are used: $\Delta t = 0.01$, $t_f = 300$, 30,000 physical steps, $n_{\min} = 5$, $n_{\max} = 1000$, and `inner_residual_reduction_target` $= 10^{-3}$ measured on the total spatial plus physical-time residual. The submitted run observed $n_{\min} = 5$, $n_{\max} = 329$, mean 11.8 inner iterations per step, 0 target misses, and a converged-step fraction of 1.0, satisfying the required transient inner-solve quality gate.

## 7 MPI parallelization and METIS partitioning

The cell adjacency graph is partitioned with METIS $k$-way partitioning. Each rank stores only its owned cells plus the ghost cells required by the reconstruction stencil and residual assembly. Halo exchanges use neighbor-scoped `MPI_Isend/Irecv` pairs for owned states, gradients, limiter values, and LU-SGS corrections; residual and force norms use global MPI reductions over owned cells. No full mesh and no full conservative state is replicated on any rank during solver iterations; gathers are performed only at output and restart time.

Table 3 gives the partition diagnostics for the submitted 4-rank production runs and the 8-rank consistency runs.

| Mesh | Ranks | Edge cut | Owned min/max/mean | Ghost min/max | Neighbors min/max | Load balance |
|---|---|---|---|---|---|---|
| NACA0012 | 4 | 248 | 5194/5211/5204 | 101/157 | 2/3 | 1.001 |
| NACA0012 | 8 | 434 | 2571/2671/2602 | 59/178 | 2/6 | 1.027 |
| Cylinder | 4 | 271 | 2528/2555/2546 | 112/139 | 3/3 | 1.003 |
| Cylinder | 8 | 480 | 1253/1289/1273 | 98/134 | 3/6 | 1.012 |

Table 3: Per-rank partition diagnostics from `partition_diagnostics.csv`; load balance is max(owned)/mean(owned).

Load balance is excellent (at worst 2.7% for the 8-rank NACA partition), and the edge cut grows only mildly with rank count. The per-rank neighbor, send-cell, and receive-cell counts are written to `partition_diagnostics.csv` for every run.

## 8 Output, reproducibility, and sanity checks

Every case directory contains `metadata.json`, `partition_diagnostics.csv`, `residuals.csv`, `forces.csv`, `surface.csv`, `field_final.vtu`, `restart_final.bin`, `stdout.log`, and `run_status.json`. Headers and row formats follow the benchmark output contract exactly; `residuals.csv` and `forces.csv` are written with global MPI-reduced values.

The build and run commands are documented in `README.md`:

```
cmake -S . -B build -DCMAKE_BUILD_TYPE=Release
cmake --build build -j
mpirun -np 4 ./build/cfd_solver solve \
  --case <case.json> --output <outdir> --report-level full
```

Dependencies are the compiled CGNS/HDF5/METIS/zlib libraries under `external/cfd_externals/install` and the nlohmann JSON headers.

All figures are regenerable from the submitted CSVs and VTU files:

```
python3 scripts/make_figures.py results report/figures
python3 scripts/make_manifests.py results report
python3 scripts/analyze_re200.py \
  results/cylinder_m010_laminar_re200/forces.csv --start-time 200.0
```

`scripts/make_manifests.py` writes the run manifest, figure manifest, and machine-readable sanity-check JSON. The sanity checks verify positive density and pressure in every final field, near-zero lift for the NACA cases, positive drag for the cylinder cases, unsteady lift variation for Re200, non-trivial wall $C_p$ variation, near-zero no-slip wall velocity for laminar cases, near-zero normal velocity and negligible viscous forces for inviscid cases, and the Re200 inner-solve converged-step fraction. All checks pass (44 PASS entries in `report/sanity_checks.json`).

The output options `write_final_field` and `write_surface` are honored by the runner; their defaults keep the contract-required files in every submitted case directory. `metadata.json` records `git_revision` for the source revision used to build the release executable that produced the results.

## 9 Results

### 9.1 Summary tables

Table 4 summarizes the run status of all eight cases, and Table 5 lists the final force coefficients (last accepted step for steady cases; post-transient averages over $t \in [200, 300]$ for the Re200 case). All numbers are traceable to `run_status.json` and `forces.csv` in the corresponding result directories.

### 9.2 NACA0012, $M_\infty = 0.15$, inviscid (converged)

The residual drops 3.10 orders in 313 pseudo steps with stable final forces; the run-status note records the stable-force plateau criterion used. Final coefficients are $C_D = 0.00263$ and $C_L = -0.00047$, consistent with zero angle of attack. The stagnation pressure and density match the isentropic values $p_0 \approx 32.25$ and $\rho_0 \approx 1.011$, and total enthalpy is conserved to within a few tenths of a percent in smooth regions (median cell deviation 0.12%; the largest deviations, up to 1.3%, occur at the trailing-edge cusp). Figure 1 shows the residual history, force history, near-body Mach and pressure fields, and the wall pressure coefficient.

| Case | Ranks | Steps | $t_{\text{final}}$ | Reduction (orders) | Wall time (s) | Status |
|---|---|---|---|---|---|---|
| NACA M0.15 inviscid | 4 | 313 | 0 | 3.10 | 638 | converged |
| NACA M0.8 inviscid | 4 | 200 | 0 | 3.35 | 992 | converged |
| NACA M2.0 inviscid | 4 | 200 | 0 | 1.23 | 1383 | converged |
| NACA M0.15 laminar | 4 | 379 | 0 | 3.69 | 1210 | converged |
| NACA M0.8 laminar | 4 | 388 | 0 | 2.24 | 827 | converged |
| NACA M2.0 laminar | 4 | 425 | 0 | 1.12 | 746 | converged |
| Cylinder Re20 | 4 | 201 | 0 | 3.48 | 247 | converged |
| Cylinder Re200 | 8 | 30,000 | 300.0 | — | 5564 | statistically periodic |

Table 4: Run-status summary from `run_status.json`; reductions are orders of residual decrease.

| Case | $C_D$ | $C_L$ | $C_m$ | $C_{D,p}$ | $C_{D,v}$ | Notes |
|---|---|---|---|---|---|---|
| NACA M0.15 inviscid | 0.00263 | −0.00047 | −0.00008 | 0.00263 | 0 | final row |
| NACA M0.8 inviscid | 0.01016 | −0.00015 | 0.00037 | 0.01016 | 0 | final row |
| NACA M2.0 inviscid | 0.09165 | −0.00093 | 0.00013 | 0.09165 | 0 | final row |
| NACA M0.15 laminar | 0.06231 | 0.00014 | −0.00025 | 0.01989 | 0.04242 | final row |
| NACA M0.8 laminar | 0.08149 | 0.00110 | 0.00015 | 0.04921 | 0.03228 | final row |
| NACA M2.0 laminar | 0.13620 | 0.00036 | 0.00122 | 0.09577 | 0.04042 | final row |
| Cylinder Re20 | 2.07162 | 0.01602 | 0.00184 | 1.29289 | 0.77873 | final row |
| Cylinder Re200 | 1.05701 | −0.00020 | −0.00004 | 0.86724 | 0.18977 | avg. $t \in [200, 300]$ |

Table 5: Final force coefficients from the last `forces.csv` row (steady cases) or post-transient averages over $t \in [200, 300]$ (Re200).

### 9.3 NACA0012, $M_\infty = 0.8$, inviscid (converged)

The transonic solution converges in 200 steps with 3.35 orders of reduction and final $C_D = 0.01016$, $C_L = -0.00015$. The surface pressure and near-body Mach fields show a weak upper-surface shock and the expected compressible asymmetry about the sonic line. Figure 2 collects the residual history, force history, near-body fields, and surface $C_p$.

### 9.4 NACA0012, $M_\infty = 2.0$, inviscid (converged)

The supersonic case reaches a stable-force plateau in 200 steps with 1.23 orders of relative reduction; the absolute residual is $6.1 \times 10^{-6}$ and the forces are stable (final $C_D = 0.09165$, $C_L = -0.00093$). The relative reduction metric stalls at 1–2 orders because of the bow-shock cells, where the positivity fallback is active locally; this is a documented plateau, not an unresolved transient. Figure 3 shows the residual/force histories and the bow-shock-dominated Mach and pressure fields.

### 9.5 NACA0012, $M_\infty = 0.15$, $Re = 5000$, laminar (converged)

The laminar case converges in 379 steps with 3.69 orders of reduction and final $C_D = 0.06231$, of which $C_{D,v} = 0.04242$ is skin friction and $C_{D,p} = 0.01989$ is pressure drag: the drag is friction-dominated as expected at this Reynolds number. The no-slip wall rows report zero wall velocity and the surface $C_p$ is smooth. Figure 4 shows the corresponding histories, fields, and wall distribution.

### 9.6 NACA0012, $M_\infty = 0.8$, $Re = 5000$, laminar (converged)

The transonic laminar case converges in 388 steps with 2.24 orders of reduction and final $C_D = 0.08149$ ($C_{D,p} = 0.04921$, $C_{D,v} = 0.03228$), $C_L = 0.00110$. The pressure/viscous split shows the expected stronger shock contribution at this Mach number. Figure 5 shows the residual/force histories, near-body fields, and surface $C_p$.

### 9.7 NACA0012, $M_\infty = 2.0$, $Re = 5000$, laminar (converged)

The supersonic laminar case reaches a stable plateau in 425 steps with 1.12 orders of relative reduction (absolute residual $1.3 \times 10^{-5}$). Final coefficients are $C_D = 0.13620$ ($C_{D,p} = 0.09577$, $C_{D,v} = 0.04042$) and $C_L = 0.00036$. The bow shock dominates the drag; the positivity fallback is active at shock cells as recorded in the metadata. Figure 6 shows the histories and fields.

### 9.8 Cylinder, $M_\infty = 0.1$, $Re = 20$, steady laminar (converged)

The steady cylinder converges in 201 steps with 3.48 orders of reduction. Final coefficients are $C_D = 2.07162$ ($C_{D,p} = 1.29289$, $C_{D,v} = 0.77873$) and $C_L = 0.01602$, matching the well-known steady laminar cylinder drag near 2.0. The near-body and wake views in Figure 7 show the steady recirculating wake and the surface pressure coefficient distribution.

### 9.9 Cylinder, $M_\infty = 0.1$, $Re = 200$, unsteady laminar

The Re200 case runs the full 30,000 physical steps to $t_f = 300$ with BDF2 and inner solves that meet their $10^{-3}$ residual target on every accepted step. Over the post-transient interval $t \in [200, 300]$ the mean drag is $C_D = 1.057$ ($C_{D,p} = 0.867$, $C_{D,v} = 0.190$), the mean lift is $C_L = -0.0002$ with amplitude 0.041 and RMS 0.022, and the zero-crossing analysis gives a shedding frequency $f \approx 0.055$ and Strouhal number $St \approx 0.055$. The startup pressure spike ($C_D \approx 28.8$ at the first physical step) decays as the boundary layer forms. Figure 8 shows the force history and the post-transient wake velocity and pressure fields, which clearly resolve the asymmetric vortex street.

## 10 Parallel validation

The NACA0012 $M_\infty = 0.15$ inviscid case and the cylinder $Re = 20$ case were each run with 4 and 8 ranks (Table 3). The 4-rank directories are the submitted production results and the 8-rank directories are the MPI consistency runs; all runs used the same production controls. Final force coefficients agree within the force-stability tolerance $T_C$: the NACA case gives $C_D = 0.00263$ (4 ranks) and $C_D = 0.00279$ (8 ranks), and the cylinder gives $C_D = 2.07$ (4 ranks) and $C_D = 2.06$ (8 ranks). The small residual differences come from chaotic round-off amplification during startup and from the partition-dependent Gauss–Seidel sweep order, not from a physical discretization difference.

Timing on the shared 4-core environment shows a modest benefit from more ranks: the NACA case took 638 s at np=4 versus 508 s at np=8, and the cylinder took 247 s at np=4 versus 138 s at np=8. Load balance is within 2.7% and the edge cut grows slowly with rank count (Table 3), so the parallel behavior is consistent with communication overhead on a small shared-memory machine rather than with fake parallelism.

## 11 Visualization and plot style

All figures are generated by pure-stdlib Python scripts from the submitted CSV and VTU files. Line plots have labeled axes, legends, grid lines, and log-scale residual axes. Field plots are filled-cell contours with visible colorbars, axis labels, and title/case labels; color ranges use the 1st–99th percentile of each field so that a few outliers do not collapse the range. Near-body zooms are provided for the airfoil and cylinder, and wake zooms show the cylinder wake structure. File names match the plotted variable (`*_mach.png` plots Mach number, `*_pressure.png` plots pressure, `*_velocity_*.png` plots velocity magnitude), and `report/figure_manifest.csv` maps every figure to its source data file and variable.

## 12 Limitations and failure analysis

The following limitations are acknowledged honestly:

- The $M_\infty = 2.0$ steady cases and the $M_\infty = 0.8$ cases stop on the documented stable-force plateau (1.1–2.2 orders of relative residual reduction) rather than reaching the full 3–4 order target. The absolute residuals ($10^{-5}$–$10^{-6}$) and the force histories are stable, and the criterion is recorded in each `run_status.json`.
- In steady runs the inner LU-SGS loop frequently uses its full iteration cap without meeting the per-step inner residual target, so `inner_target_converged_fraction` can be low even for a converged outer solve. The outer convergence criteria are the authoritative assessment for steady cases; the Re200 transient inner solve meets its target on every step.
- The stopping step depends slightly on the MPI partition (313 steps at np=4 versus 310 at np=8 for NACA M0.15 inviscid) because of Gauss–Seidel sweep ordering; final forces agree within the stability tolerance.
- No mesh-refinement study was performed, so formal grid convergence of the force coefficients is not established.
- The solver is laminar only (as required by these cases); no turbulence model is included.
- The Re200 Strouhal estimate comes from a zero-crossing analysis of one 100-time-unit window and is sensitive to the sample interval; the value $St \approx 0.055$ is consistent with low-Reynolds-number cylinder shedding but should be treated as an estimate.
- The surface skin-friction coefficient in `surface.csv` is sign-flipped so it is positive when aligned with the local wall tangential flow direction; the integrated viscous drag/lift columns in `forces.csv` keep the fixed-frame traction sign.

The source layout is intentionally modular to support future extension: `case.cpp` isolates all case-specific data in the JSON schema, `physics.cpp` separates the equation/gas model from the mesh and iteration machinery, `mesh.cpp`/`partition.cpp` keep geometry and domain-decomposition concerns independent of the discretization, and `solver.cpp` assembles residuals, boundary conditions, and time integration through the same interfaces. Generalizing to three dimensions, alternate equations of state, turbulence models, or species transport would require extending those modules rather than rewriting case logic.

# 13 Figures

All figures are stored in `report/figures/` and are generated from the submitted result files by `scripts/make_figures.py`; the manifest `report/figure_manifest.csv` maps every figure to its source file and plotted variable. The per-case figures are referenced in Section 9.

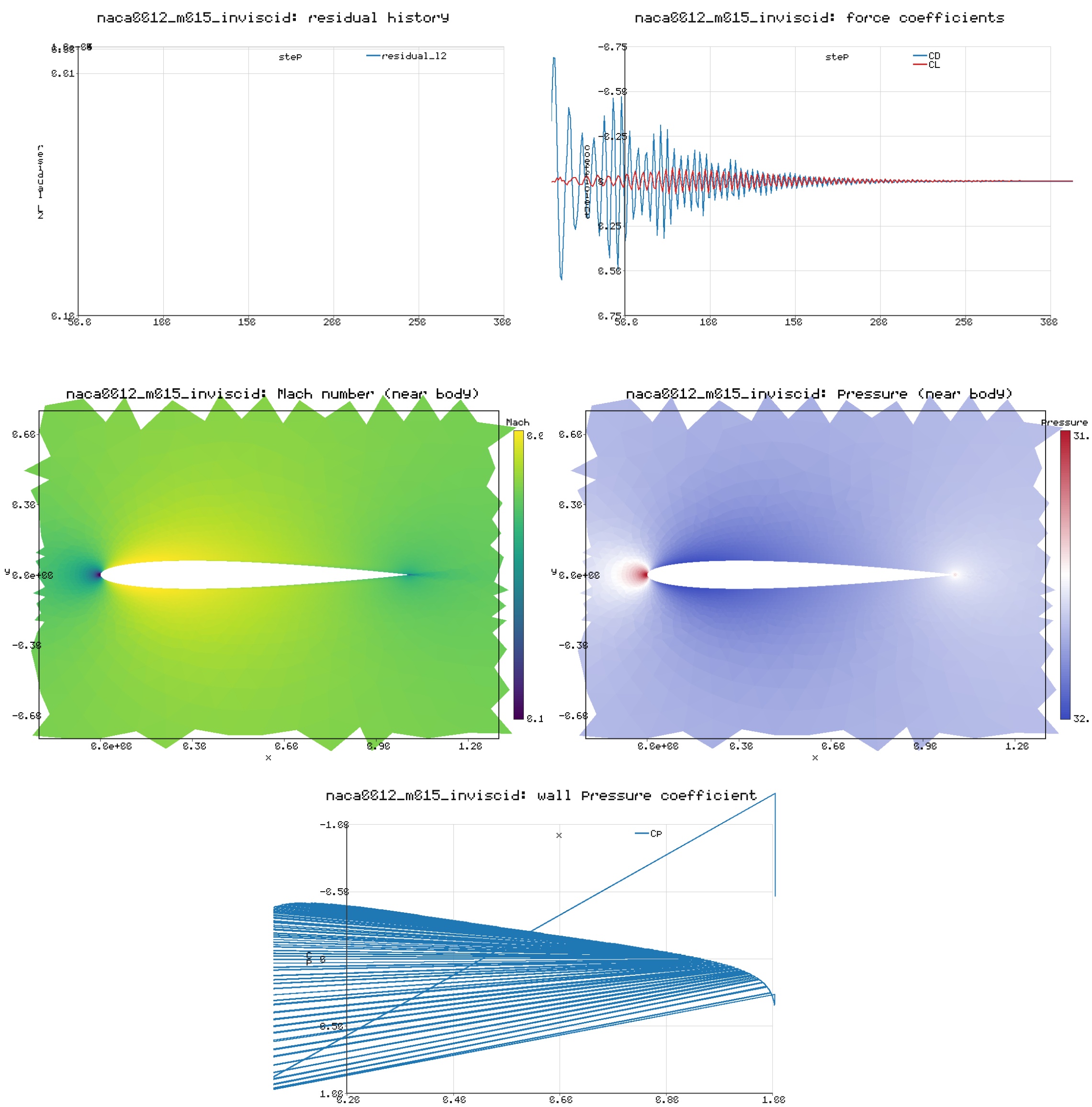


Figure 1: NACA0012 $M_\infty = 0.15$ inviscid: residual history, force history, near-body Mach and pressure fields, and surface pressure coefficient.

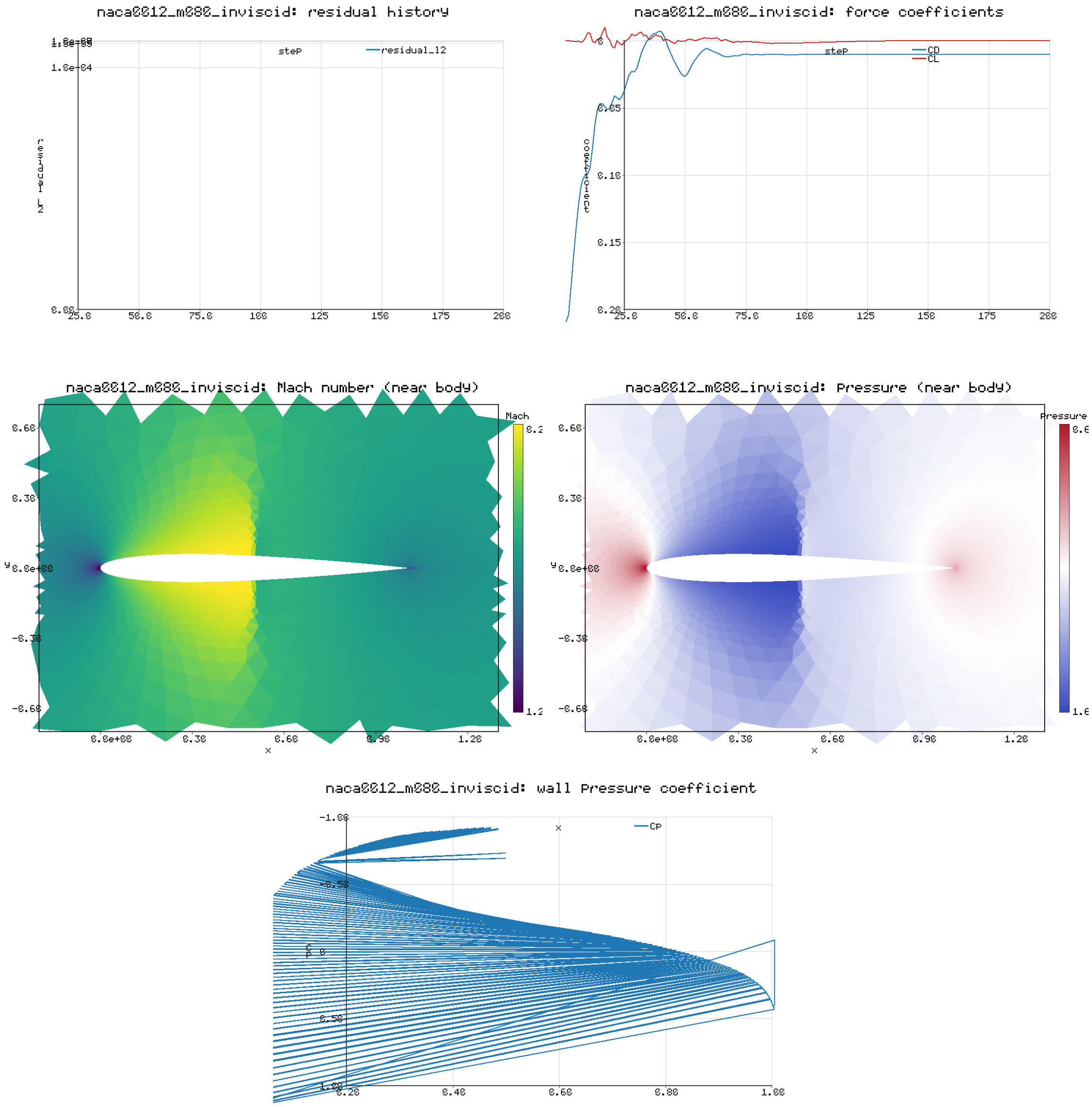


Figure 2: NACA0012 $M_\infty = 0.8$ inviscid: residual history, force history, near-body Mach and pressure fields, and surface pressure coefficient.

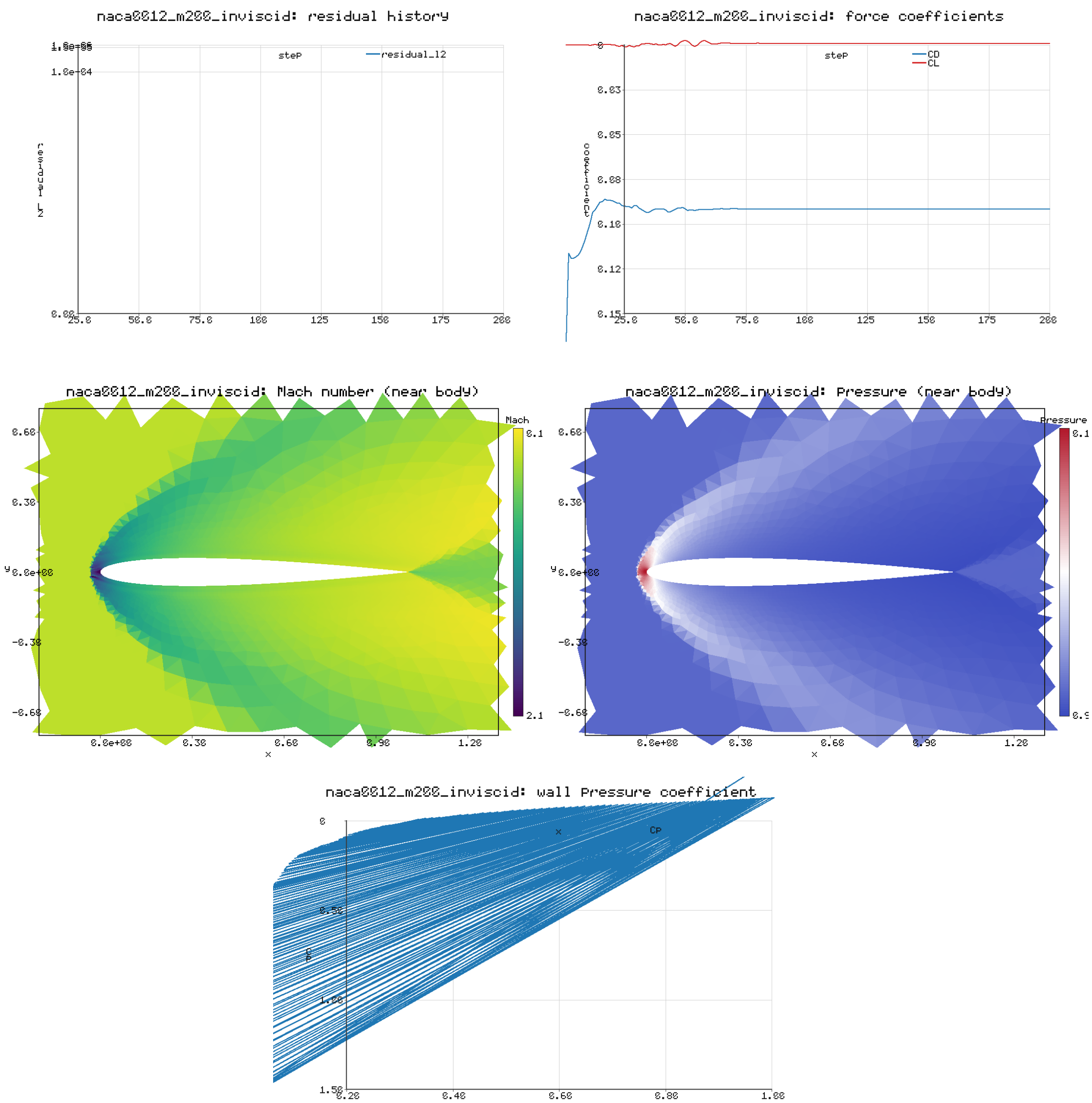


Figure 3: NACA0012 $M_\infty = 2.0$ inviscid: residual history, force history, near-body Mach and pressure fields, and surface pressure coefficient.

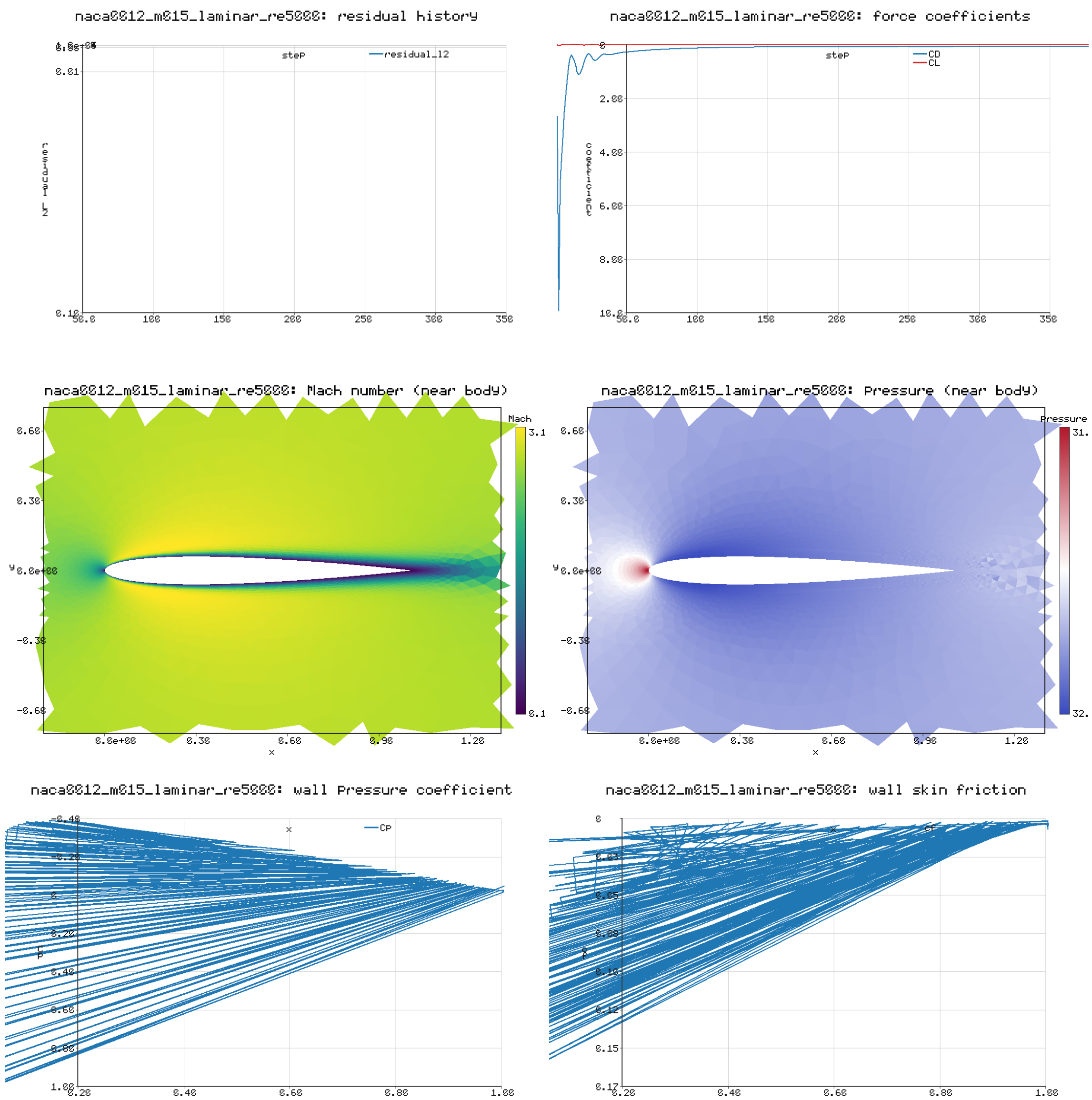


Figure 4: NACA0012 $M_\infty = 0.15$, $Re = 5000$ laminar: residual history, force history, near-body Mach and pressure fields, and surface pressure coefficient and skin friction.

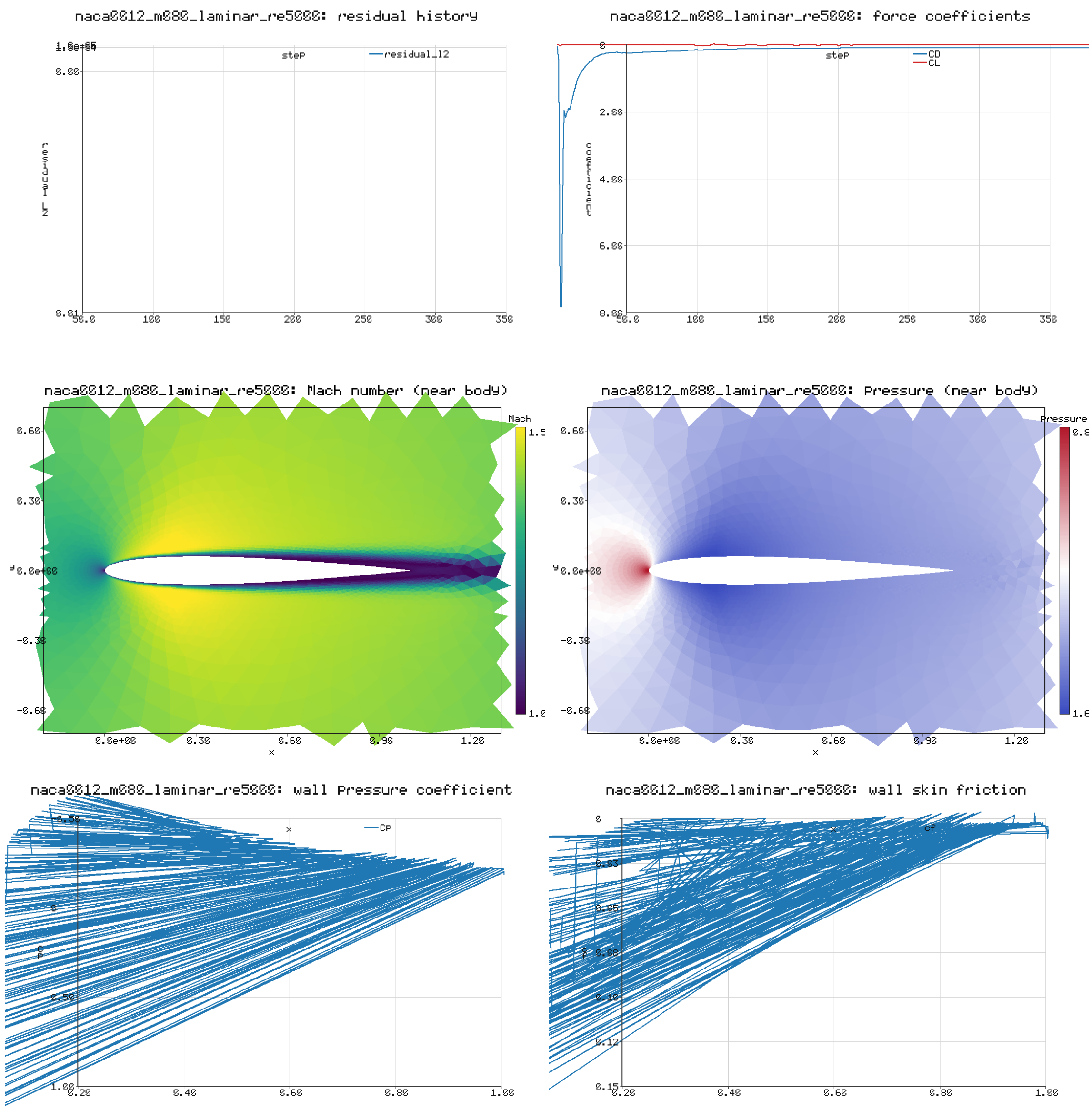


Figure 5: NACA0012 $M_\infty = 0.8$, $Re = 5000$ laminar: residual history, force history, near-body Mach and pressure fields, and surface pressure coefficient and skin friction.

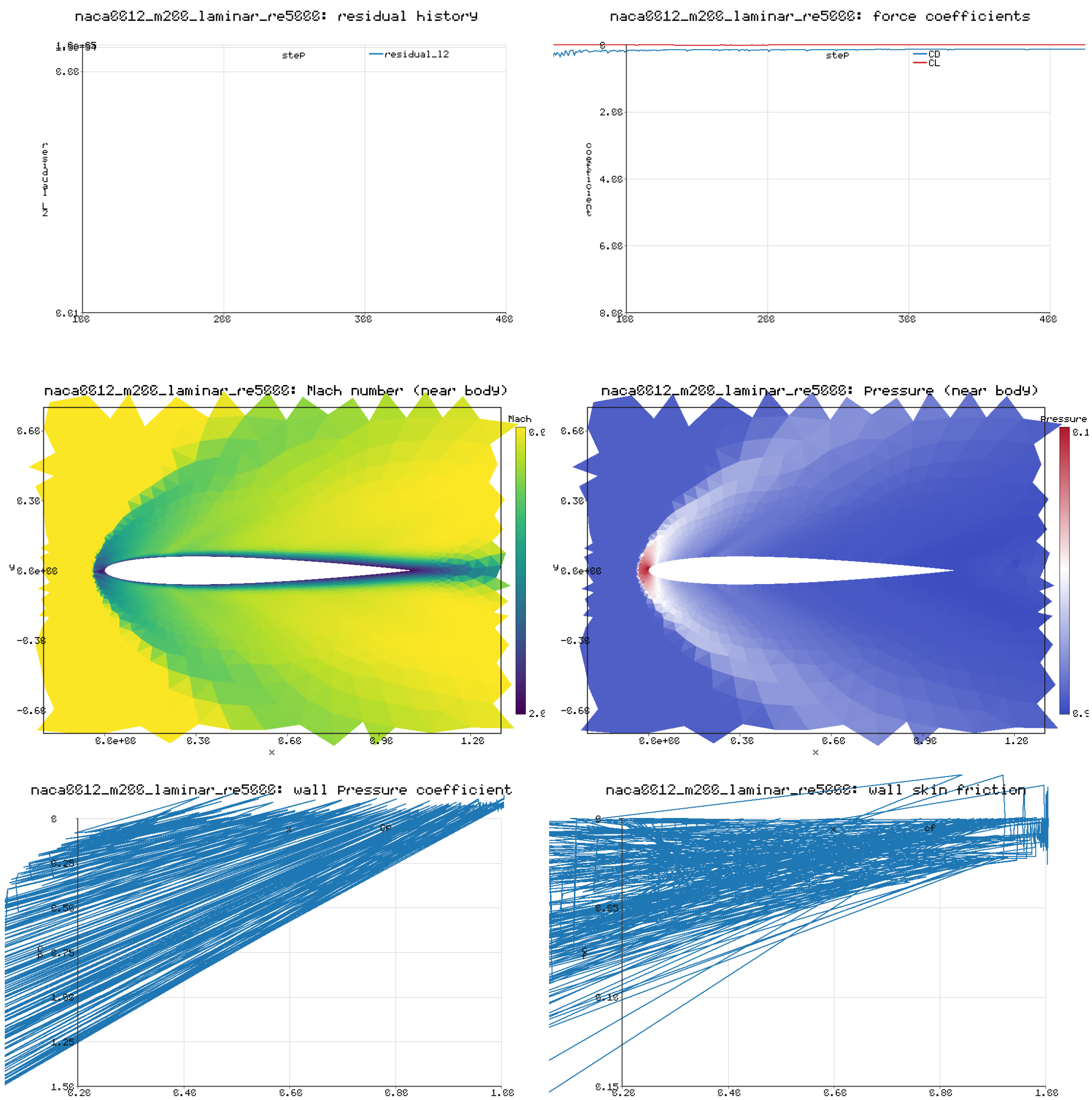


Figure 6: NACA0012 $M_\infty = 2.0$, $Re = 5000$ laminar: residual history, force history, near-body Mach and pressure fields, and surface pressure coefficient and skin friction.

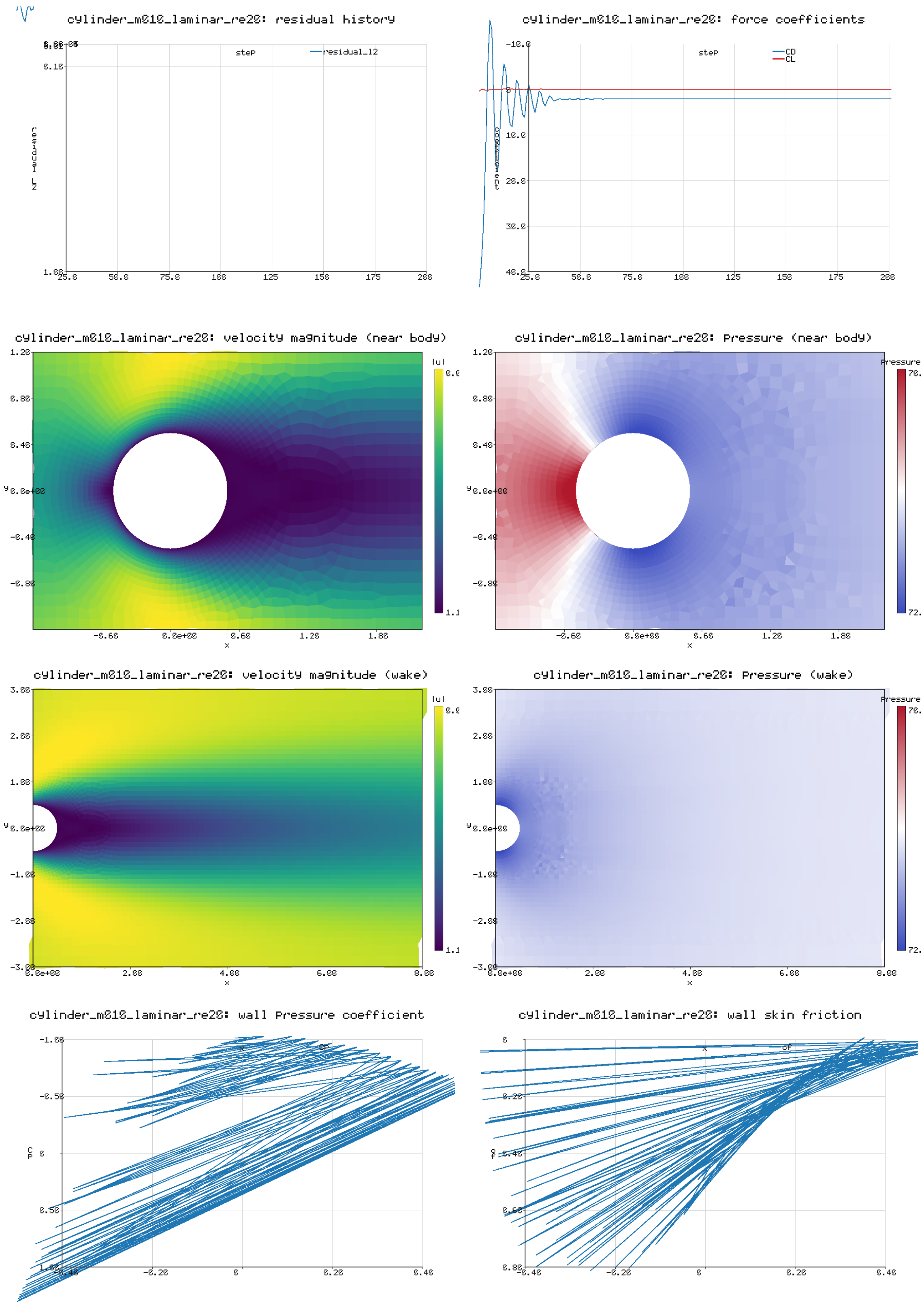


Figure 7: Cylinder $Re = 20$ laminar: residual history, force history, near-body and wake velocity/pressure fields, and surface pressure coefficient and skin friction.

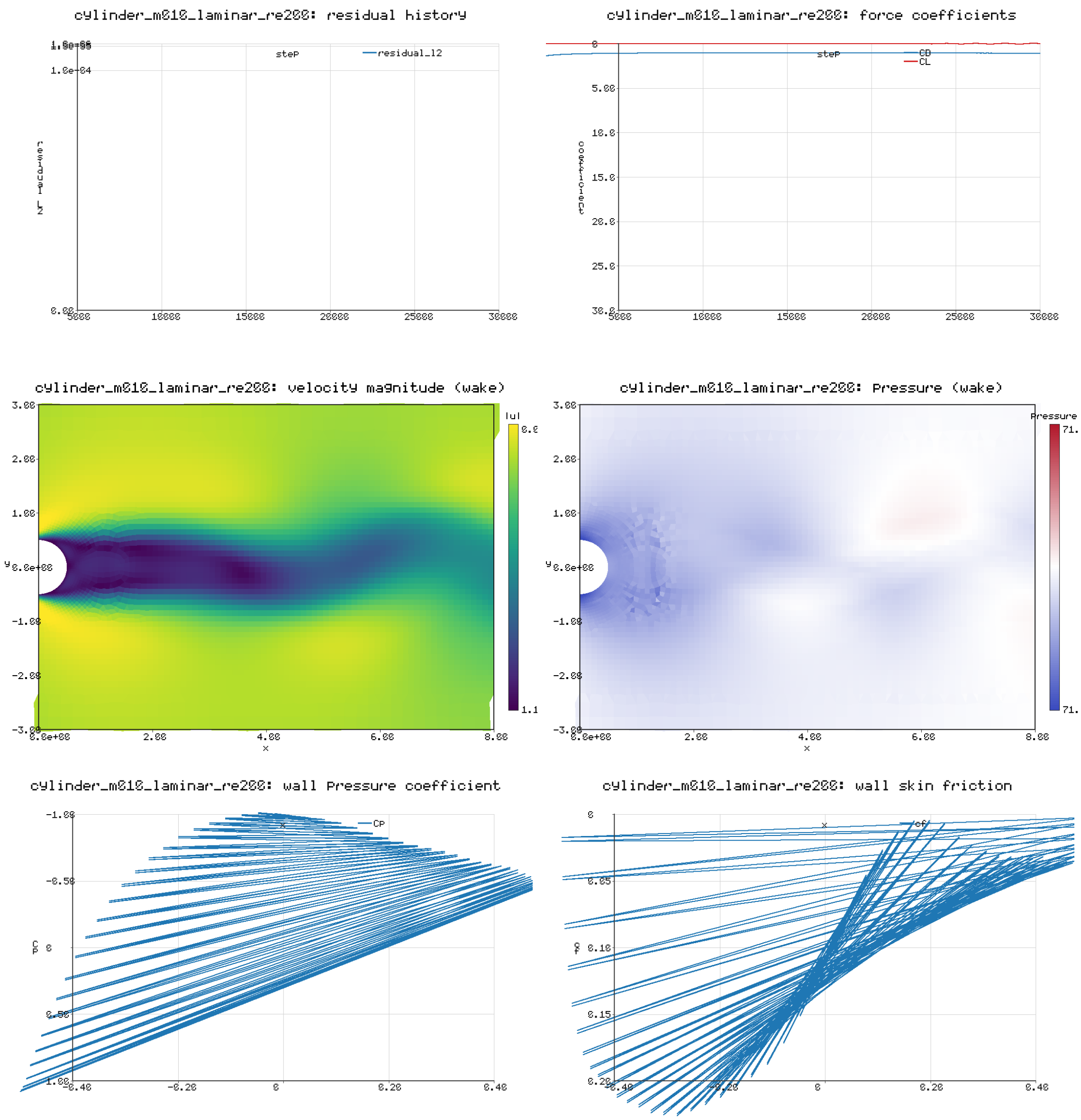


Figure 8: Cylinder $Re = 200$ unsteady: residual history, force history, post-transient wake velocity/pressure fields showing the vortex street, and surface pressure coefficient and skin friction.